

August 14, 2026

Luca Bacco

AI RESEARCH SCIENTIST

Rome, Italy

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Profile

Assistant professor and **biomedical engineer** with specialized expertise in **AI** and **NLP**, focusing on developing innovative solutions to support digital health applications, interpreting AI behaviors under constraints, and integrating multiple modalities. Demonstrated experience in analyzing heterogeneous datasets, including time series, imaging, and textual data, with over 30 publications in **high-impact international journals** and **conferences**.

Highly proficient and experienced with diverse **programming languages**, frameworks, and databases, used to develop AI/NLP pipelines, web-based platforms, and data integration systems. Skilled in rapidly adapting to new technologies and methodological settings, with strong **versatility** and problem-solving abilities across biomedical AI, multimodal learning, generative AI evaluation, and cross-domain applications.

Experienced in supervising and **mentoring** students and research teams, actively fostering **interdisciplinary collaboration** within national and international research projects. Expertise in developing **AI-driven healthcare solutions**, ranging from clinical NLP applications and generative AI evaluation to multimodal analysis of biosignals, medical imaging, and clinical data.

Fluent in **Italian** and professionally proficient in **English**, with significant international academic experience in scientific writing, presentations, and teaching.

Academia

Since September 2025, I have held an **Assistant Professor** position (Ricercatore RTDA) at the Università Campus Bio-Medico di Roma, where I also serve as the internationalization coordinator for the master's program in Biomedical Engineering. Before this appointment, I was awarded two research fellowships and a scholarship. My academic path builds on a PhD in Science and Engineering for Humans and the Environment (Information Technology for Biomedicine curriculum), and includes an international research period as a Visiting PhD Student at the University of Groningen.

Università Campus Bio-Medico di Roma, Department of Engineering

ASSISTANT PROFESSOR (RICERCATORE RTDA)

Rome, Italy

Sep. 2025 - present

Università Campus Bio-Medico di Roma, Department of Engineering

POSTDOCTORAL RESEARCHER (2 ASSEGNI DI RICERCA + 1 BORSA DI STUDIO)

Rome, Italy

Nov. 2023 - Aug. 2025

Università Campus Bio-Medico di Roma, Department of Engineering

PHD CANDIDATE (XXXV CYCLE)

Rome, Italy

Nov. 2019 - Nov. 2022

University of Groningen, Center of Language and Cognition

VISITING PHD STUDENT

Groningen, Netherlands

Jan. 2022 - July 2022

AFFILIATIONS

Research Unit of Intelligent Health Technology for Health and Wellbeing

DEPARTMENT OF ENGINEERING, UNIVERSITÀ CAMPUS BIO-MEDICO DI ROMA

Rome, Italy

2025 - present

ItaliaNLP Lab

INSTITUTE OF COMPUTATIONAL LINGUISTICS "A.ZAMPOLLI" OF THE ITALIAN NATIONAL RESEARCH COUNCIL (ILC-CNR)

Pisa, Italy

2019 - present

Research Unit of Computer Systems and Bioinformatics

DEPARTMENT OF ENGINEERING, UNIVERSITÀ CAMPUS BIO-MEDICO DI ROMA

Rome, Italy

2019 - 2025

PROJECT PARTICIPATION

Throughout my academic career, I have actively contributed to several collaborative projects involving universities, research institutes, and companies, while consistently maintaining a clear research focus on AI and its real-world applications.

GRACE

AI RESEARCHER IN BIOMEDICINE

Sep. 2025 - present

Contributing to the design of AI-driven decision-support solutions for heart failure care, focusing on the integration of digitized clinical guidelines, explainable ML models, and clinical data within evidence-based care pathways.

XAI-CARE

AI RESEARCHER IN BIOMEDICINE

Jan. 2023 - Feb. 2026

Contributed to scientific writing activities and to the collection and organization of clinical and multimodal data for the development of computer vision, NLP, and multimodal AI models aimed at patient characterization and decision-support applications. **Technical skills:** computer vision, multimodality, phenotyping. **Transversal skills:** scientific writing, interdisciplinary communication, clinical data management.

Digital Innovation in Price Statistics

AI RESEARCHER IN ECONOMICS

Mar. 2024 - Aug. 2025

Collected, structured, and organized heterogeneous data sources for data-driven pipelines aimed at improving the timeliness and scalability of price statistics. Conducted applied research on AI-based time series methods. **Technical skills:** time series analysis.

Domain skills: economic and statistical measurement understanding

DRIVERS

FULL-STACK DEVELOPER

Nov. 2022 - Mar. 2024

Built a full-stack web platform from scratch to support interpretable cyber and climate risk assessment for industrial facilities.

Technical skills: ASP.NET, C#, JavaScript, MongoDB, API integration, web scraping. **Transversal skills:** collaboration with experts outside my core research domain, acquisition of domain-specific requirements.

DHelp4H

NLP EARLY-STAGE RESEARCHER

Nov. 2019 - Oct. 2022

Bridged academic research and industrial development by translating NLP methods into practical tools. **Technical skills:** NLP, Python (Hugging Face, Tensorflow, PyTorch, sklearn, pandas). **Transversal skills:** project coordination, interdisciplinary collaboration.

PATENT APPLICATIONS

Artificial intelligence system for non-invasive detection of calcium levels via ECG

Rome, Italy

PATENT APPLICATION NO 2502168.4

13 February 2025

Education

Università Campus Bio-Medico di Roma

Rome, Italy

PHD IN *Science and Engineering for Humans and the Environment - Information Technology for Biomedicine*

March 2023

Thesis: *Exploring New Technologies in Healthcare: Advancing Natural Language Processing* **Brief description:** The increasing digitalization of healthcare has made clinical and biomedical documents a crucial source of information, opening new opportunities for Natural Language Processing. In this context, the thesis explores the latest NLP methods and Transformer-based technologies within a sensitive and rapidly evolving healthcare domain, focusing on explainability, medical text simplification, and the reliability of language models. **Supervisors:** Felice Dell'Orletta (ItaliaNLP Lab, ILC-CNR), Mario Merone (IHT, Università Campus Bio-Medico di Roma)

Università Campus Bio-Medico di Roma

Rome, Italy

MASTER'S DEGREE IN BIOMEDICAL ENGINEERING (110/110 CUM LAUDE)

February 2019

Thesis: *Development of a motion control system for an anthropomorphic manipulator based on online extraction of muscle synergies.*

Università degli Studi di Roma Tor Vergata

Rome, Italy

BACHELOR'S DEGREE IN MEDICAL ENGINEERING

March 2017

CERTIFICATIONS

License to Practice as an Industrial Engineer

Rome, Italy

July 2020

Research

My scientific production includes 23 journal papers and 10 international conference papers, together with 8 conference papers accepted for presentation this year. I actively contributed to all these works, as reflected by the high number of first, co-first, and corresponding authorships across most of them.

My publications trace the evolution of a research agenda focused on understanding how AI models behave under domain-specific constraints and varying data quality conditions, how generative models can be fairly evaluated, and how heterogeneous information sources can be integrated to build more trustworthy and holistic AI systems. Most of this work is grounded in biomedical applications, with a focus on interpretable decision support and on NLP/LLM-based methods whose outputs must be evaluated beyond fluency or surface-level correctness.

BIBLIOMETRIC INDEX

Scopus	h-index: 10	Citations: 293	www.scopus.com/authid/detail.uri?authorId=57220927387
Google Scholar	h-index: 12	Citations: 436	https://scholar.google.com/citations?user=Aw4cEzMAAAAJ

SELECTED JOURNAL PUBLICATIONS

- Giaccone, P.*, D'Antoni, F.*, **Bacco, L.***, Russo, F., Papalia, G.F., Vadalà, G., Papalia, R., Pecchia, L., Denaro, V., Merone, M. "Artificial intelligence based intervertebral discs grading: From unsupervised detection to explainable Pfirrmann classification". **Biomedical Signal Processing and Control**. (2026)
- Piperno, R.*, **Bacco, L.***, Dell'Orletta, F., Merone, M., Pecchia, L. "Cross-Lingual Distillation for Domain Knowledge Transfer with Sentence Transformers". **Knowledge-Based Systems**. (2025)
- Bonfigli, A., **Bacco, L.†**, Pecchia, L., Merone, M., Dell'Orletta, F. "Efficient Multi-Task Learning with Instance Selection for Biomedical NLP". **Computers in Biology and Medicine**. (2025)

4. Petrosino, L., **Bacco, L.[†]**, Salvati, G., Merone, M., Papi, M. "A GARCH-temporal fusion transformer model for the volatility prediction of exchange traded funds". **Neural Computing and Applications**. (2025)
5. Piperno, R., **Bacco, L.**, Petrosino, L., Matarrese, M., Merone, M., Pecchia, L. "Feature-Based Knowledge Distillation for Explainable Detection of Pulmonary Diseases". **Health and Technology**. (2025)
6. Bonfigli, A.* , **Bacco, L.***, Merone, M., Dell'Orletta, F. "From Pre-training to Fine-tuning: an in-depth Analysis of Large Language Models in the Biomedical Domain". **Artificial Intelligence in Medicine**. (2024)
7. **Bacco, L.**, Petrosino, L., Arganese, D., Vollero, L., Papi, M., Merone, M. "Investigating Stock Prediction Using LSTM Networks and Sentiment Analysis of Tweets Under High Uncertainty: A Case Study of North American and European Banks". **IEEE Access**. (2024)
8. **Bacco, L.**, Dell'Orletta, F., Lai, H., Merone, M. Nissim, M. "A Text Style Transfer System for Reducing the Physician-Patient Expertise Gap: an Analysis with Automatic and Human Evaluations". **Expert Systems With Applications**. (2023)
9. **Bacco, L.**, Minnema, G., Caselli, T., Dell'Orletta, F., Merone, M., Nissim, M. "On the Instability of Further Pre-Training: Does a Single Sentence Matter to BERT?". **Natural Language Processing Journal**. (2023)
10. d'Angelis, O.* , **Bacco, L.***, Vollero, L., Merone, M. "Advancing ECG Biometrics through Vision Transformers: A Confidence-Driven Approach". **IEEE Access**. (2023)
11. Farinella, F., Merone, M., **Bacco, L.**, Capirchio, A., Ciccozzi, M., Caligiore, D. "Machine Learning analysis of High-Grade Serous Ovarian Cancer proteomic dataset reveals novel candidate biomarkers". **Scientific Reports**. (2022)
12. **Bacco, L.**, Cimino, A., Merone, M., Dell'Orletta, F. "Explainable Sentiment Analysis: A Hierarchical Transformer-Based Extractive Summarization Approach". **Electronics**. (2021)

SELECTED CONFERENCE WORKS

1. Della Morte, D., Merone, M., Pecchia, L., **Bacco, L.** "On the Evaluation of Biomedical Text Simplification Datasets: A Benchmarking Study". **XVII Mediterranean Conference on Medical and Biological Engineering and Computing**. (Medicon 2026)
2. Bonfigli, A., Merone, M., Dell'Orletta, F., ... **Bacco, L.** "Benchmarking Automatic Question Generation from Clinical Practice Guidelines for Education: a Case Study for Low Back Pain Management". **XVII Mediterranean Conference on Medical and Biological Engineering and Computing**. (Medicon 2026)
3. Ferraro, C., Pasquali, F., **Bacco, L.[†]**, Merone, M., Pecchia, L. "ECG-Based Non-Invasive Glycaemic Estimation in Type 1 Diabetes". **XVII Mediterranean Conference on Medical and Biological Engineering and Computing**. (Medicon 2026)
4. Bonfigli, A., Di Giacinto, A.M., Giordani, F., Pecchia, L., Merone, M., **Bacco, L.[†]** "A Multimodal Wearable Framework for Predicting Epileptic Seizures". **24th International Conference on Artificial Intelligence in Medicine**. (AIME 2026)
5. Piperno, R., **Bacco, L.[†]**, Papalia, G.F., Russo, F., Maltese G., Vadalà, G., Papalia, R., Pecchia, L., Merone, M. "Multi-View Lumbar MRI for Therapy-Oriented Decision Support System in Low Back Pain". **24th International Conference on Artificial Intelligence in Medicine**. (AIME 2026)
6. Piperno, R., Bonfigli, A., Dell'Orletta, F., Pecchia, L., Merone, M., **Bacco, L.[†]** "Doctor, Is That You? Evaluating Large Language Models on Italy's Medical School Entrance Exams". **11th Italian Conference on Computational Linguistics**. (CLiC-it 2025)
7. Bonfigli, A., Piperno, R., **Bacco, L.[†]**, Dell'Orletta, F., et al. "A Novel Real-World Dataset of Italian Clinical Notes for NLP-based Decision Support in Low Back Pain Treatment". **11th Italian Conference on Computational Linguistics**. (CLiC-it 2025)
8. **Bacco, L.**, Cimino, A., Merone, M., Dell'Orletta, F. "Extractive Summarization for Explainable Sentiment Analysis using Transformers". **18th Extended Semantic Web Conference**. (ESWC 2021)

* , Co-first authors

[†], Corresponding author

Teaching

My teaching activity complements my research profile and reflects my commitment to transferring methodological, technical, and applied knowledge to students. Across lectures, seminars, and student supervision, I have contributed to educational activities related to AI, programming, data analysis, and biomedical engineering.

Università Campus Bio-Medico di Roma

LECTURER

- Advanced Processing and AI for Biomedical Data, Signals and Images - Master's Degree Course in *Biomedical Engineering* - 9 CFU
- Elaborazione Avanzata ed AI per Dati, Segnali ed Immagini Biomedicali - Master's Degree Course in *Ingegneria Biomedica* - 9 CFU
- Applications of GenAI and Deep Learning for Health and Wellbeing - Master's Degree Course in *Biomedical Engineering* - 6 CFU

2025/26

Institute of Cognitive Sciences and Technologies of the Italian National Research Council (ISTC-CNR) - AI2Life srl

LECTURER

- Machine Learning Advanced - *Advanced School in Artificial Intelligence* - 8h
- Machine Learning II - *Advanced School in Artificial Intelligence* - 8h

2024/25

2023/24

CEDEL ELIS college - Polytechnic University of Milan

LECTURER

- Basi di Dati 2 - Bachelor's Degree Course in *Ingegneria Digitale* - 9h

2020/21

Università Campus Bio-Medico di Roma

TEACHING ASSISTANT

- Biomedical Signal Processing - Bachelor's Degree Course in *Biomedical Engineering* - 11h
- Signal and Image Processing - Bachelor's Degree Course in *Medtech* - 11h
- Deep Learning for Big Data - Master's Degree Course in *Intelligent Systems Engineering* - 12h
- Deep Learning for Big Data - Master's Degree Course in *Intelligent Systems Engineering* - 12h
- Informatica - Bachelor's Degree Course in *Fisioterapia* - 8h
- Deep Learning for Big Data - Master's Degree Course in *Intelligent Systems Engineering* - 4 CFU
- Elaborazione dei Segnali e delle Immagini Digitali - Master's Degree Course in *Ingegneria Biomedica* - 3 CFU
- Informatica - Bachelor's Degree Course in *Fisioterapia* - 8h
- Deep Learning for Big Data - Master's Degree Course in *Intelligent Systems Engineering* - 4 CFU
- Elaborazione dei Segnali e delle Immagini Digitali - Master's Degree Course in *Ingegneria Biomedica* - 3 CFU
- Informatica - Bachelor's Degree Course in *Fisioterapia* - 8h
- Informatica Medica - Single-cycle Master's Degree Course in *Medicina e Chirurgia* - 4 CFU
- Sistemi Informatici e Igiene - Bachelor's Course in *Tecniche di Radiologia Medica per Immagini e Radioterapia* - 1 CFU

2024/25

2023/24

2022/23

2021/22

2020/21

Supervision

Since the academic year 2019/20, my academic activity has included the supervision of 5 theses and the co-supervision of 15 theses, including 17 master's theses in Biomedical Engineering (BE) and Intelligent Systems Engineering (ISE), and 3 bachelor's theses in Industrial Engineering (IE). These experiences have allowed me to guide research-oriented work, support the development of technical and methodological skills, and involve students in topics related to AI, NLP, biomedical data analysis, and digital health applications.

Università Campus Bio-Medico di Roma

THESIS SUPERVISOR (RELATORE)

- **Master's Degree in BE:** *Subject-Independent Stress Detection through Adversarial Training*
- **Master's Degree in BE:** *Physiology-Conditioned Generative AI for Synthetic Echocardiographic Video Synthesis*
- **Master's Degree in BE:** *Design and Development of a Multimodal Wearable Framework for Predicting Epileptic Seizures*
- **Master's Degree in ISE:** *From Sign Language To Speech: Development of a Pipeline for Expressive Speech Synthesis*
- **Bachelor's Degree in IE:** *Design and Development of a System for Automatic Creation and Analysis of Google Forms for Complex Clinical Questionnaires*

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Università Campus Bio-Medico di Roma

THESIS ADVISOR (CORRELATORE)

- **Master's Degree in BE:** *Development of a Subject-Specific Adaptive System for Non-Invasive Blood Glucose Estimation from ECG*
- **Master's Degree in BE:** *Developing a Literature-Informed Chatbot to support Research on Nanoparticle-Based breast cancer treatments*
- **Master's Degree in BE:** *A Novel Drawdown-informed Framework using Deep Learning and Reinforcement Learning for Optimized Biomedical ETF Investments*
- **Master's Degree in BE:** *Automatic Cardiac Risk Management Classification using Electronic Health Record*
- **Master's Degree in ISE:** *Use of Large Language Models to improve Cloud Infrastructure Defense*
- **Master's Degree in ISE:** *A Framework for Tracking Price Trends in the Online Grocery Marketplace: from Automated Data Mining to Business Intelligence*
- **Master's Degree in ISE:** *Improving The Efficiency Of Aeronautical Communications: Design And Development Of A Speech Recognition System For Air Traffic Control*
- **Master's Degree in BE:** *A Practical Approach to Device-aware Pruning*
- **Master's Degree in BE:** *Innovative Approaches to Explainability Distillation: Methodologies and Debates*
- **Master's Degree in ISE:** *Pre-Training and Fine-Tuning in the Biomedical Domain: Exploring and Adapting Large Language Models*
- **Master's Degree in BE:** *Development of an Algorithm for Photoplethysmographic Signal Analysis: Quality Control and Blood Pressure Prediction*
- **Bachelor's Degree in IE:** *Postural Data Analysis using AI-powered Classification Models*
- **Master's Degree in BE:** *Application of Process Mining techniques for the analysis of nursing staff processes within the UCBM Polyclinic*
- **Bachelor's Degree in IE:** *Designing a Process Mining Application for Patient Monitoring*
- **Master's Degree in BE:** *Design and development of an ECG identification system*

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2019/20

Outside Academia

This section includes selected experiences beyond my core academic appointments, highlighting scientific and professional activities, industry-oriented teaching, international exposure, oral presentations, and post-graduate training.

SCIENTIFIC ACTIVITIES

I aim to maintain continuous engagement in peer-review activities for international journals and conferences, supporting the eval-

uation and dissemination of high-quality research in my research field. Past and current reviewing activities include international journals such as Biomedical Signal Processing and Control, Expert Systems with Applications, Artificial Intelligence in Medicine, IEEE Access, among others, and conferences such as ACL, CBMS, MetroInd4.0&IoT, CLiC-it, and LREC.

OTHER WORK EXPERIENCES

BPCOmedia S.r.l.

STARTUP AMBASSADOR + FULL-STACK DEVELOPER

Rome, Italy / London, UK

May 2019 - Nov. 2019

Startup Ambassador at the *Global Startup Program*, in collaboration with the *Italian Trade Agency (ICE/ITA)* and *TechItalia:Lab*: presented the startup at international events, promoted its value proposition to diverse stakeholders, and generated qualified contacts with potential investors and partners. **Full-Stack Developer**: implemented new functionalities within a web platform for patients and physicians, performed reverse engineering activities, and integrated front-end and back-end components. **Technical skills**: C#, ASP.NET, TypeScript, AngularJS. **Transversal skills**: pitch delivery, cross-cultural communication, presentation design.

OTHER TEACHING AND TALKS

EOS Management Consulting

LECTURER

- Pacchetto Office e Principi di Intelligenza Artificiale - Consiglio Superiore della Magistratura (PerForma PA) - 24h

2025/26

Università Campus Bio-Medico di Roma

LECTURER

- Let's pAI: Discover how AI learns from the world around us - Summer School UCBM for high school students - 2h

2025/26

UCBM Academy

LECTURER

- Intelligenza Artificiale per l'uomo e per l'impresa - Unindustria - 2h
- Fondamenti di Programmazione e Linguaggio C - I level Master in *Cybersecurity Management* - 6h

2024/25

PERIODS ABROAD

University of Groningen

VISITING PHD STUDENT @ CENTER OF LANGUAGE AND COGNITION (CLCG), COMPUTATIONAL LINGUISTICS DEPARTMENT

Groningen, Netherlands

Jan. 2022 - July 2022

During my visiting period, I conducted research under the supervision of Prof. Malvina Nissim and collaborated with researchers (T. Caselli) and PhD students (H. Lai, G. Minnema), contributing to the works listed as [8] and [9] in the selected journal publications.

TechItalia:Lab

STARTUP AMBASSADOR @ GLOBAL STARTUP PROGRAM, IN COLLABORATION WITH ITALIAN TRADE AGENCY (ITA)

London, UK

Oct. 2019 - Nov. 2019

ORAL PRESENTATIONS

Abstract	The Impact of Data Size and Quality on Biomedical NLP , Rome Technopole 2024	Rome, Italy
Abstract	Unveiling a New Resource for Crossing Linguistic Frontiers in Biomedicine , EMBEC 2024	Portoroz, Slovenia
Paper	Natural Language Processing in Healthcare: a Bird's Eye View , AI*IA 2023	Rome, Italy
Paper	Postural Data Analysis using AI-powered Classification Models , METROIND4.0&IOT 2023	Brescia, Italy
Paper	Photoplethysmographic Signal Quality Assessment , METROIND4.0&IOT 2023	Brescia, Italy
Paper	Extractive Summarization for Explainable Sentiment Analysis , ESWC 2021	Hersonissos, Greece (Online)
Paper	Machine Learning for Sentiment Analysis for Italian Reviews in Healthcare , CLiC-it 2020	Bologna, Italy (Online)

POST-MASTER ACTIVITIES

2024	International Summer School on Wearable Sensors , Certificate of attendance	Rome, Italy
2023	AILC Lectures , Certificate of attendance	Pisa, Italy
2022	Natural Language Processing , University of Groningen - Best student project award	Groningen, Netherlands
2022	Research Seminar Information Science , University of Groningen	Groningen, Netherlands
2022	4th Advanced Course of Data Science & Machine Learning (ACDL) , Certificate of attendance	Certosa di Pontignano, Italy
2021	Memory Networks , AIDA course - Certificate of attendance	Online
2020	AI-enabled health care: from decision support to autonomous robots , XXXIX Scuola Annuale di Bioingegneria, Gruppo Nazionale di Bioingegneria (GNB) - Certificate of attendance	Online
2019	Natural Language Processing con Python: il Corso Pratico , Udemy - Certificate of completion	Online
2019	Design Thinking Bootcamp for students , University of Rome La Sapienza - Certificate of attendance	Rome, Italy
2019	Global Startup Program (Special Edition) 2019 , TechItalia:Lab - Certificate of completion	London, UK

PARTICIPATION IN INTERNATIONAL CONFERENCES

2026	24th International Conference on Artificial Intelligence in Medicine (AIME) , Certificate of attendance	<i>Ottawa, Canada</i>
2024	9th European Medical and Biological Engineering Conference (EMBEC) , Certificate of attendance	<i>Ljubljana, Slovenia</i>
2023	22nd International Conference of the Italian Association for Artificial Intelligence (AIxIA) , Certificate of attendance	<i>Rome, Italy</i>
2023	IEEE MetroInd4.0&IoT , Certificate of attendance	<i>Brescia, Italy</i>
2021	18th European Semantic Web Conference (ESWC) , Certificate of attendance	<i>Hernossinos, Greece (online)</i>
2021	7th Italian Conference on Computational Linguistic (CLiC-it) ,	<i>Bologna, Italy (online)</i>

Skills

Languages	Italian (native) English
Programming	Proficiency with Python and C# Hands-on experience with JavaScript, HTML, CSS, C/C++, MATLAB, SQL
Frameworks	Hugging Face, TensorFlow & Keras, PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, seaborn ASP.NET
Databases	Hands on experience with MongoDB and RavenDB
Others	LaTeX Linux GitHub Office (Word , Excel , PowerPoint) Canva Web Scraping
Soft skills	Interdisciplinary communication and collaboration Mentoring Critical thinking Analytical mindset Continuous learning Versatility Academic writing Public speaking Teamwork

Aware that false declarations entail the application of the criminal sanctions provided for in art. 76 of D.P.R. 445/2000, I declare that the above information corresponds to the truth and I authorize the processing of my personal data as per the regulations in force (Legislative Decree 196/2003, containing the Code for the Protection of Personal Data, and EU Regulation no. 679/2016 GDPR).

Rome, August 14, 2026

List of Publications (Complete)

JOURNAL PUBLICATIONS

2026

23. Giaccone, P.*, D'Antoni, F.*, **Bacco, L.***, Russo, F., Papalia, G.F., Vadalà, G., Papalia, R., Pecchia, L., Denaro, V., Merone, M. "Artificial intelligence based intervertebral discs grading: From unsupervised detection to explainable Pfirrmann classification". **Biomedical Signal Processing and Control**.
22. Sassi, M., Mancini, L., Carnevale, A., Mancuso, M., Giaccone, P., **Bacco, L.**, et al. "Preoperative Radiomic Biomarkers Reflect Functional Shoulder Impairment in Rotator Cuff Tears: A Structure-Function Analysis". **Frontiers in Bioengineering and Biotechnology**.
21. Muraro, M.*, Ligan, J.*, **Bacco, L.*†**, et al. "How technologies are driving digital innovation in mosquito surveillance systems: a global scoping review over the past decade". **BMJ Public Health**.
20. Longo, U.G., Giaccone, P., **Bacco, L.**, et al. "Discriminative and predictive power of radiomics features from preoperative MRI studies in patients with supraspinatus lesions". **BMC Musculoskeletal Disorders**.

2025

19. Crispo A., Pagnano, M.E., Bonfigli, A., Pecchia, L., Luongo, A., Porciello, G., Prete, M., Coluccia, S. **Bacco, L.**, et al. "Adopting Machine Learning to Predict Breast Cancer Patients Adherence with Lifestyle Recommendations and Quality of Life Outcomes". **Frontiers in Digital Health**.
18. Sasso, D., **Bacco, L.†**, Palumbo, L., Marcucci, J., Salvini, N., Laureti, T., Vollero, L. "Price Variations in Food Products: A Time Series Dataset for Analysis Across Regions in Italian Supermarkets". **Data in Brief**.
17. Petrosino, L., **Bacco, L.†**, Salvati, G., Merone, M., Papi, M. "A GARCH-temporal fusion transformer model for the volatility prediction of exchange traded funds". **Neural Computing and Applications**.
16. Bonfigli, A., **Bacco, L.†**, Pecchia, L., Merone, M., Dell'Orletta, F. "Efficient Multi-Task Learning with Instance Selection for Biomedical NLP". **Computers in Biology and Medicine**.
15. Chiorboli, S. D'Ambrosio, A., Mazzeo, L., Santori, F., **Bacco, L.**, D'Antoni, F., Palombo, G., Merone, M., Piemonte, V. "A Prediction of Drugs Transport, Distribution and Absorption by a Multicompartmental Physiologically-Based Pharmacokinetic Model". **Engineering Proceedings**.
14. Piperno, R., **Bacco, L.**, Petrosino, L., Matarrese, M., Merone, M., Pecchia, L. "Feature-Based Knowledge Distillation for Explainable Detection of Pulmonary Diseases". **Health and Technology**.
13. Piperno, R.*, **Bacco, L.***, Dell'Orletta, F., Merone, M., Pecchia, L. "Cross-Lingual Distillation for Domain Knowledge Transfer with Sentence Transformers". **Knowledge-Based Systems**.

2024

12. Bonfigli, A.*, **Bacco, L.***, Merone, M., Dell'Orletta, F. "From Pre-training to Fine-tuning: an in-depth Analysis of Large Language Models in the Biomedical Domain". **Artificial Intelligence in Medicine**.
11. Tamburella, F., Lorusso, M., Merone, M., **Bacco, L.**, Molinari, M., Tramontano, M., Scivoletto, G., Tagliamonte, N.L. "Quantifying Treatments as Usual and with Technologies in Neurorehabilitation of Individuals with Spinal Cord Injury". **Healthcare**.
10. **Bacco, L.**, Petrosino, L., Arganese, D., Vollero, L., Papi, M., Merone, M. "Investigating Stock Prediction Using LSTM Networks and Sentiment Analysis of Tweets Under High Uncertainty: A Case Study of North American and European Banks". **IEEE Access**.

2023

9. **Bacco, L.**, Dell'Orletta, F., Lai, H., Merone, M. Nissim, M. "A Text Style Transfer System for Reducing the Physician-Patient Expertise Gap: an Analysis with Automatic and Human Evaluations". **Expert Systems With Applications**.
8. **Bacco, L.**, Minnema, G., Caselli, T., Dell'Orletta, F., Merone, M., Nissim, M. "On the Instability of Further Pre-Training: Does a Single Sentence Matter to BERT?". **Natural Language Processing Journal**.
7. d'Angelis, O.*, **Bacco, L.***, Vollero, L., Merone, M. "Advancing ECG Biometrics through Vision Transformers: A Confidence-Driven Approach". **IEEE Access**.
6. D'Antoni, F., Petrosino, L., Marchetti, A., **Bacco, L.**, Pieralice, S., Vollero, L., Pozzilli, P., Piemonte, V., Merone, M. "Layered meta-learning Algorithm for Predicting Adverse Events in Type 1 Diabetes". **IEEE Access**.

2022

5. **Bacco, L.**, Russo, F., Ambrosio, L., D'Antoni, F., Vollero, L., Vadalà, G., Dell'Orletta, F., Merone, M., Papalia, R., Denaro, V. "Natural language processing in low back pain and spine diseases: A systematic review". **Frontiers in Surgery**.
4. D'Antoni, F., Russo, F., Ambrosio, L., **Bacco, L.**, Vollero, L., Vadalà, G., Merone, M., Papalia, R., Denaro, V. "Artificial Intelligence and Computer Aided Diagnosis in Chronic Low Back Pain: A Systematic Review". **International Journal of Environmental Research and Public Health**.
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MANUSCRIPTS UNDER REVIEW

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